United Nations

Our Common Future
Chapter 5: Food Security: Sustaining The Potential

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1. The world produces more food per head of population today than ever before in human history. In 1985, it produced nearly 500 kilogrammes per head of cereals and root crops, the primary sources of food./1 Yet amid this abundance, more than 730 million people did not eat enough to lead fully productive working lives./2 There are places where too little is grown; there are places where large numbers cannot afford to buy food. And there are broad areas of the Earth, in both industrial and developing nations, where increases in food production are undermining the base for future production.

2. The agricultural resources and the technology needed to feed growing populations are available. Much has been achieved over the past few decades. Agriculture does not lack resources; it lacks policies to ensure that the food is produced where it is needed and in a manner that sustains the livelihoods of the rural poor. He can meet this challenge by building on our achievements and devising new strategies for sustaining food and livelihood security.

I. Achievements

3. Between 1950 and 1985, cereal production outstripped population growth, increasing from around 700 million tons to over 1,800 million tons, an annual growth rate of around 2.7 per cent./3 This increase helped to meet escalating demands for cereals caused by population growth and rising incomes in developing countries and by growing needs for animal feed in developed countries. Yet regional differences in performance have been large. (See Table 5-1.)

4. As production has increased sharply in some regions and demand in others, the pattern of world trade in foods, especially cereals, has changed radically. North America exported barely 5 million tons of foodgrains yearly before the Second World War; it exported nearly 120 million tons during the 1980s. Europe's grain deficit is very much lower now. and the bulk of North American exports are to the USSR. Asia, and Africa. Three countries - China, Japan, and the USSR - took half the world exports in the early 1980s; much of the rest went to relatively wealthy developing countries, such as Middle Eastern oil exporters. Several poor agricultural countries, especially in sub-Saharan Africa, have become net importers of foodgrains. Still, although one-fourth of sub-Saharan Africa's population relied on imported grains in 1984, that region's imports have accounted for less than 10 per cent of world grain trade thus far in the 1980s./4

Table 5-1

Two Decades of Agricultural Development

	Per Capita Food Production (Index 1961-64 = 100)		Per Capita Gross Cropped Area (Hectares)		Per Hectare Fertilizer use (kg.)	
	1961-64	1981-84	1964	1984	1964	1984
World	100	112	0.44	0.31	29.3	85.3
North America	100	121	1.05	0.90	47.3	93.2
Western Europe	100	131	0.11	0.25	114.4	124.1
Eastern Europe and USSR	100	128	0.84	0.71	30.4	122.1
Africa	100	88	0.74	0.35	1.8	9.7
Near East*	100	107	0.53	0.35	6.9	53.6
Far East**	100	116	0.10	0.20	4.4	45.4

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Latin America	100	108	0.49	0.45	11.6	11.4
CPE's of Asia***	100	135	0.17	0.10	15.5	170.3

* An FAO grouping that includes West Asia plus Egypt, Libya and Sudan

** An FAO grouping that covers South and South-East Asia excluding the centrally planned economies of Asia.

*** An FAO grouping of Centrally Planned Economies of Asia which covers China, Kampuchea, North Korea. Mongolia and Vietnam.

5. Other foods besides grains are changing the patterns of world food demand and production. Demand for milk and meat is growing as incomes rise in societies that prefer animal protein. and much agricultural development in the industrialized nations has been devoted to meeting these demands. In Europe, meat production more than tripled between 1950 and 1984, and milk production nearly doubled./5 Meat production for exports increased sharply, particularly in the rangelands of Latin America and Africa. World meat exports have risen from around 2 million tons in 1950-52 to over 11 million tons in 1984./6

6. To produce this milk and meat required in 1984 about 1.4 billion cattle and buffaloes, 1.6 billion sheep and goats, 800 million pigs, and a great deal of poultry - all of which weigh more than the people on the planet./7 Most of these animals graze or browse or are fed local plants collected for them. However, rising demands for livestock feedgrains led to sharp increases in the production of cereals such as corn, which accounted for nearly two-thirds of the total increase in grain production in North America and Europe between 1950 and 1985.

7. This unprecedented growth in food production has been achieved partly by an extension of the production base: larger cropped areas, more livestock, more fishing vessels, and so on. But most of it is due to a phenomenal rise in productivity. Population increases have meant a decline in the area of cropped land in most of the world in per capita terms. And as the availability of arable land has declined, planners and farmers have focused on increasing productivity. In the past 35 years this has been achieved by:

- using new seed varieties designed to maximize yields, facilitate multiple cropping, and resist disease;
- applying more chemical fertilizers, the consumption of which rose more than ninefold/8;
- using more pesticides and similar chemicals, the use of which increased thirty-two-fold/9; and
- increasing irrigated area, which more than doubled./10

8. Global statistics mask substantial regional differences. (See Box 5-1.) The impacts of new technology have been uneven, and in some respects the agricultural technology gap has widened. For instance, average African foodgrain productivity declined in relation to European productivity from roughly one-half to about one-fifth over the past 35 years. Even in Asia, where new technology has spread rapidly, productivity in relation to European levels dropped./11 Similar 'technology gaps' have emerged between regions within countries.

9. The past few decades have seen the emergence of three broad types of food production systems, 'Industrial agriculture', capital- and input-intensive and usually large-scale, is dominant in North America, Western and Eastern Europe, Australia and New Zealand, and in some small areas in developing countries. 'Green Revolution agriculture' is found in uniform, resource-rich, often flat and irrigated areas in the agricultural heartlands of some developing countries. It is more widespread in Asia but is also found in parts of Latin America and North Africa. Though initially the new technologies may have favoured large farmers, they are today accessible to a growing number of small producers. 'Resource-poor agriculture' relies on uncertain rain rather than irrigation and is usually found in developing regions difficult to farm - drylands, highlands, and forests - with fragile soils. This includes most of sub-Saharan Africa and the remoter areas of Asia and Latin America. Here, per capita production has been declining and hunger is a critical problem. But today, all three systems of food production display signs of crises that endanger their growth.

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II. Signs of Crisis

10. Agricultural policies in practically all countries have focused on output growth. Despite this, it has proved far more difficult to raise world agricultural output by a consistent 3 per cent a year in the mid-1980s than it was in the mid-1950s. Moreover, production records have been offset by the appearance

Box 5-1 Regional Perspectives on Agricultural Development

Africa

- a drop in per capita food output of about 1 per cent a year since the beginning of the 1970s
- a focus on cash crops and a growing dependence on imported food, fostered by pricing policies and foreign exchange compulsions
- major gaps in infrastructure for research, extension, input supply, and marketing
- degradation of the agricultural resource base due to desertification, droughts, and other processes
- large untapped potential of arable land, irrigation, and fertilizer use

West Asia and North Africa

- improvements in productivity due to better irrigation, the cultivation of high-yielding varieties, and higher fertilizer use
- limited arable land and considerable amounts of desert, making food self-sufficiency a challenge
- a need for controlled irrigation to cope with dry conditions **South and East Asia**
- increased production and productivity, with some countries registering grain surpluses
- rapid growth in fertilizer use in some countries and extensive development of irrigation
- government commitments to be self-reliant in food, leading to national research centres, development of high-yielding seeds, and the fostering of location-specific technologies
- little unused land, and extensive, unabated deforestation
- growing numbers of rural landless

Latin America

- declining food imports since 1980, as food production kept pace with population growth over the last decade
- government support in the form of research centres to develop high-yielding seeds and other technologies
- inequitable distribution of land
- deforestation and degradation of the agricultural resource base, fueled partly by foreign trade and debt crisis
- a huge land resource and high productivity potential, though most of the potentially arable land is in the remote, lightly populated Amazon Basin, where perhaps only 20 per cent of the land is suitable for sustainable agriculture

North America and Western Europe

- North America the world's leading source of surplus foodgrain, though the rate of increase in output per hectare and in total productivity slowed in the 1970s

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- subsidies for production that are ecologically and economically expensive
- depressing effect of surpluses on world markets and consequent impact on developing countries
- a resource base increasingly degraded through erosion, acidification, and water contamination
- in North America, some scope for future agricultural expansion in frontier areas that can be intensively farmed only at high cost

Eastern Europe and the Soviet Union

- food deficits met through imports, with the Soviet Union being the world's largest grain importer
- increased government investment in agriculture accompanied by eased farm distribution and organization to meet desires for food self-reliance, leading to production increases in meat and root crops
- pressures on agricultural resources through soil erosion. acidification, salinization, alkalization, and water contamination of linked economic and ecological crises: Industrialized countries are finding it increasingly difficult to manage their surplus food production, the livelihood base of millions of poor producers in developing countries is deteriorating, and the resource base for agriculture is under pressure virtually everywhere.

1. Impact of Subsidies

11. The food surpluses in North America and Europe result mainly from subsidies and other incentives that stimulate production even in the absence of demand. Direct or indirect subsidies, which now cover virtually the entire food cycle, have become extremely expensive. In the United States, the cost of farm support has grown from $2.7 billion in 1980 to $25.8 billion in 1986. In the EEC, such costs have risen from $6.2 billion in 1976 to $21.5 billion in 1986./12

12. It has become politically more attractive, and usually cheaper, to export surpluses - often as food aid - rather than to store them. These heavily subsidized surpluses depress the international market prices of commodities such as sugar and have created severe problems for several developing countries whose economies are based on agriculture. Non-emergency food aid and low-priced imports also keep down prices received by Third World farmers and reduce the incentive to improve domestic food production.

13. The environmental consequences of a heavily subsidized production system are becoming evident within industrialized nations/13;

- lower productivity as soil quality declines due to intensive soil cultivation and overuse of chemical fertilizers and pesticides/14;
- the destruction of the countryside, through clearing of hedgerows, park: belts, and other protective cover and the levelling, occupation, and cultivation of marginal land and watershed protection areas; and
- nitrate pollution of ground-water aquifers due to the often subsidized overuse of nitrate fertilizers,

14. The financial, economic, and environmental effects of the current incentive systems are beginning to be questioned by many governments and groups, including farm organizations. A particular area of concern is the impact of these policies on developing countries. They depress international prices of products, such as rice and sugar, that are important exports for many developing countries and so reduce exchange earnings of developing countries. They increase the instability of world prices. And they discourage the processing of agricultural commodities in the producing countries./15

15. It is in the interests of all, including the farmers, that the policies be changed. Indeed, in recent years some conservation-oriented changes have taken place and some subsidy systems have increasingly stressed the need to retire land from production. The financial and economic burden of subsidies must be reduced. The

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harm that these policies do to the agriculture of developing countries by disrupting world markets must be eliminated.

2. Neglect of the Small Producer

16. The new technology behind increases in agricultural productivity requires scientific and technological skills, a system for technology extension and other services for farmers, and commercial orientation in farm management. In many' parts of Asia, in particular, small farmers have shown a remarkable capacity to use new technology once they are given incentives and adequate financial and infrastructural support. Small cash-crop farmers in Africa have demonstrated the potential of the smallholder on that continent, and in the last few years successes have been recorded in food crops also. But ecologically disadvantaged areas and land-poor rural masses have not benefited from advances in technology and will not until governments are willing and able to redistribute land and resources, and give them the necessary support and incentives.

17. Agricultural support systems seldom take into account the special circumstances of subsistence farmers and herders. subsistence farmers cannot afford the high cash outlay of modern inputs. Many are shifting cultivators who do not have a clear title to the land they use. They may plant a variety of crops on one plot to meet their own needs, and are thus unable to use methods developed for large stands of a single crop.

18. Many herders are nomadic and difficult to reach with education, advice, and equipment. They, like subsistence farmers, depend on certain traditional rights, which are threatened by commercial developments. They herd traditional breeds, which are hardy but rarely highly productive.

19. Women farmers, though they play a critical role in food production, are often ignored by programmes meant to improve production. In Latin America, the Caribbean, and Asia they form a large agricultural labour force, while most of sub-Saharan Africa's food is grown by women. Yet almost all agricultural programmes tend to neglect the special needs of women farmers.

> I think that at a forum like this there always tends to be someone standing up and saying you forgot my issue. I think my serious sensitivity to women!s role vis-a-vis the environment.
>
> Especially in Africa, I think it has been clearly stated over and again that women are responsible for between 60 to 90 per cent of the food production, processing, and marketing. No one can really address the food crisis in Africa or many of the other crises that seem to exist here without addressing the question of women, and really seeing that women are participants in decision-making processes at the very basic all the way through up the highest level.
>
> Mrs. King
> The Greenbelt Movement
> WCED Public Hearing
> Nairobi. 23 Sept 1986

3. Degradation of the Resource Base

20. Short-sighted policies are leading to degradation of the agricultural resource base on almost every continent: soil erosion in North America: soil acidification in Europe; deforestation and desertification in Asia. Africa, and Latin America; and waste and pollution of water almost everywhere. Within 40-70 years, global warming may cause the flooding of important coastal production areas. Some of these effects arise from trends in energy use and industrial production. Some arise from the pressure of population on limited resources. But agricultural policies emphasizing increased production at the expense of environmental considerations have also contributed greatly to this deterioration.

3.1 Loss of Soil Resources

21. Increases in cropped areas in recent decades have often extended cultivation in marginal lands prone to erosion. By the late 1970s, soil erosion exceeded soil formation on about a third of U.S. cropland, much of it in the midwestern agricultural heartland./16 In Canada, soil degradation has been costing farmers $1 billion a year./17 In the USSR, the extension of cultivation to the so-called Virgin Lands was a major plank of agricultural policy, but now it is believed that much of this land is marginal./18 In India, soil erosion affects 25-30 per cent of

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the total land under cultivation./19 Without conservation measures, the total area of rainfed cropland in developing countries in Asia, Africa, and Latin America would shrink by 544 million hectares over the long term because of soil erosion and degradation, according to an FAO study./20

22. Erosion makes soil less able to retain water, depletes it of nutrients, and reduces the depth available for the roots to take hold. Land productivity declines. Eroded topsoil is carried to rivers. lakes, and reservoirs, silts up ports and waterways, reduces reservoir storage capacity, and increases the incidence and severity of floods.

23. Poorly designed and implemented irrigation systems have caused waterlogging, salinization, and alkalization of soils. FAO and UNESCO estimate that as much as half the world's irrigation schemes suffer in some degree from these problems./21 These estimates indicate that some 10 million hectares of irrigated land are being abandoned each year.

24. Soil degradation erodes the overall resource base for agriculture. The loss of croplands encourages farmers to overuse the remaining land and to move into forests and onto rangelands. Sustainable agriculture cannot be based on methods that mine and deplete the soil,

3.2 Impact of Chemicals

25. Chemical fertilizers and pesticides have played a large role in production increases since the Second World War, but clear warnings have been raised against over-reliance on them. The run-off of nitrogen and phosphates from excess use of fertilizers damages water resources, and such damage is spreading.

26. Using chemicals to control insects, pests, weeds, and fungi enhances productivity, but overuse threatens the health of humans and the lives of other species. Continuing, long-term exposure to pesticide and chemical residues in food, water, and even in the air is hazardous, particularly to children. A 1983 study estimated that approximately 10.000 people died each year in developing countries from pesticide poisoning and about 400,000 suffered acutely./22 The effects are not limited to the area where pesticides are used but travel through the food chain.

27. Commercial fisheries have been depleted, bird species endangered, and insects that prey on pests wiped out. The number of pesticide-resistant insect pest species worldwide has increased and many resist even the newest chemicals. The variety and severity of pest infestations multiply, threatening the productivity of agriculture in the areas concerned.

28, The use of agricultural chemicals is not in itself harmful. In fact, the level of use is still quite low in many regions. In these areas, response rates are high and the environmental consequences of residues are not yet a problem. Hence these regions would benefit by using more agrochemicals. However, the growth in the use of chemicals tends to be concentrated precisely where they may be doing more overall harm than good.

3.3 Pressure on Forests

29. Forests are crucial for maintaining and improving the productivity of agricultural land. Yet agricultural expansion, a growing world timber trade, and woodfuel demand have destroyed much forest cover. Although this destruction has occurred worldwide, today the greatest challenge is in developing countries, particularly in tropical forests. (See Chapter 6.)

30. Growing populations and the decreasing availability of arable land lead poor farmers in these countries to seek new land in forests to grow more food. Some government policies encourage the conversion of forests to pastures and others encourage large resettlement schemes in forests. There is nothing inherently wrong with clearing forests for farming, provided that the land is the best there is for new farming, can support the numbers encouraged to settle upon it, and is not already serving a more useful function, such as watershed protection. But often forests are cleared without forethought or planning.

31. Deforestation most severely disrupts mountainous areas and upland watersheds and the ecosystems that depend on them. The uplands influence precipitation, and the state of their soil and vegetation systems influence how this precipitation is released into the streams and rivers and onto the croplands of the plains

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below. The growing numbers and growing severity of both floods and droughts in many parts of the world have been linked to the deforestation of upland watersheds./23

3.4 Advancing Deserts

32. Some 29 per cent of the earth's land area suffers slight, moderate, or severe desertification; an additional 6 per cent is classified as extremely severely desertified./24 In 1984, the world's drylands supported some 850 million people, of whom 230 million were on lands affected by severe desertification./25

33. The process of desertification affects almost every region of the globe, but it is most destructive in the drylands of South America, Asia, and Africa; for these three areas combined. 18.5 per cent (870 million hectares) of productive lands are severely desertified. Of the drylands in developing countries. Africa's Sudano-Sahelian zones and, to a lesser extent, some countries south of this zone suffer the most. In their arid and semi-arid lands are to be found 80 per cent of the moderately affected and 85 per cent of the severely affected people./26

34. Land permanently degraded to desert-like conditions continues to grow at an annual rate of 6 million hectares./27 Each year. 21 million additional hectares provide no economic return because of the spread of desertification./28 These trends are expected to continue despite some local improvements.

35, Desertification is caused by a complex mix of climatic and human effects. The human effects, over which we have more control, include the rapid growth of both human and animal populations, detrimental land use practices (especially deforestation), adverse terms of trade, and civil strife. The cultivation of cash crops on unsuitable rangelands has forced herders and their cattle onto marginal lands. The unfavourable international terms of trade for primary products and the policies of aid donors have reinforced pressures to encourage increasing cash-crop production at any cost.

36. A Plan of Action conceived by UNEP and drawn up at the 1977 UN Conference on Desertification has led to some slight, mainly local gains./29 Progress on the plan has been hampered by lack of financial support from the international community, by inadequacies of the regional organizations established to respond to the regional nature of the problem, and by the lack of involvement of grass-roots communities.

> Small farmers are held responsible for environmental destruction as if they had a choice of resources to depend on for their livelihood, when they really don't. In the context of basic survival, today's needs tend to overshadow consideration for the environmental future. It is poverty that is responsible for the destruction of natural resources, not the poor.
>
> Geoffrey Bruce
> Canadian International Development Agency
> WCED Public Hearing
> Ottawa, 26/27 May 1986

III. The Challenge

37. Food demand will increase as populations increase and their consumption patterns change. In the remaining years of this century, about 1.3 billion people will be added to the human family (see Chapter 4); rising incomes, however, may account for 30 to 40 per cent of the increased demand for food in developing countries and about 10 per cent in industrial nations./30 Thus over the next few decades, the global food system must be managed to increase food production by 3 to 4 per cent yearly.

38. Global food security depends not only on raising global production, but on reducing distortions in the structure of the world food market and on shifting the focus of food production to food-deficit countries, regions, and households. Many of the countries not growing enough food to feed themselves possess the largest remaining reservoirs of untapped agricultural resources. Latin America and sub-Saharan Africa have much unused land. although its quality and quantity vary greatly from nation to nation and much of it is ecologically vulnerable./31 The Soviet Union and parts of North America have significant amounts of frontier land suitable for agriculture; only Asia and Europe are truly land-starved.

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39. Global food security also depends on ensuring that all people, even the poorest of the poor, can get food. While on the world scale this challenge requires a reappraisal of global food distribution, the task weighs more immediately and heavily on national governments. Inequitable distribution of production assets, unemployment, and underemployment are at the heart of the problem of hunger in many countries.

40. Rapid, sound agricultural development will mean not only more food but more opportunities for people to earn money to purchase food. Thus when countries with untapped agricultural resources provide food by importing more, they are effectively importing unemployment. By the same token, countries that are subsidizing food exports are increasing unemployment in food-importing countries. This marginalizes people, and marginalized people are forced to destroy the resource base to survive. Shifting production to food-deficit countries and to the resource-poor farmers within those countries is a way of securing sustainable livelihoods.

41. Conserving the agricultural resource base and livelihood security of the poor can be mutually supportive in three ways. First, secure resources and adequate livelihoods lead to good husbandry and sustainable management. Second, they ease rural-to-urban migration, stimulate agricultural production from resources that otherwise would be underused, and reduce the need for food to be produced elsewhere. Third, by combating poverty, they help to slow population growth.

42. Shifting the focus of production to food-deficit countries will also reduce pressures on agricultural resources in the industrialized market economies, enabling them to move towards more sustainable agricultural practices. Incentive structures can be changed so that instead of encouraging overproduction, they encourage farm practices that improve soil and water quality. Government budgets will be relieved of the burdens of storing and exporting surplus products.

43. This shift in agricultural production will be sustainable only if the resource base is secure. As indicated, this is far from the case today. Thus to achieve global food security, the resource base for food production must be sustained, enhanced, and, where it has been diminished or destroyed, restored.

There are many contradictions in agricultural development. The blind imitation of models developed under different circumstances will have to give way to the realities and conditions existing in Africa. Large areas of virgin land have been opened up for export crops whose prices keeps declining. This is not in the interest of developing countries.

There are so many problems to be overcome that we forget that every problem is an opportunity to do something positive. This is an opportunity for us to think of conservation and environment in a broad educational context. In doing so, we will be able to capture the next generation and demonstrate the wonder and the benefits of the world around them.

Adolfo Mascarenhas
IUCN Harare Office
WCED Public Hearing
Harare. 18 Sept 1986

IV. Strategies for Sustainable Food Security

44. Food security requires more than good conservation programmes, which can be - and usually are - overridden and undermined by inappropriate agricultural, economic, and trade policies. Nor is it just a matter of adding an environmental component to programmes. Food strategies must take into account all the policies that bear upon the threefold challenge of shifting production to where it is most needed, of securing the livelihoods of the rural poor, and of conserving resources.

1. Government Intervention

45. Government intervention in agriculture is the rule in both industrial and developing countries, and it is here to stay. Public investment in agricultural research and extension services, assisted farm credit and marketing services, and a range of other support systems have all played parts in the successes of the last half-century. In fact, the real problem in many developing countries is the weakness of these systems.

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46. Intervention has taken other forms as well. Many governments regulate virtually the entire food cycle - inputs and outputs, domestic sales, exports, public procurement, storage and distribution, price controls and subsidies - as well as imposing various land use regulations: acreage, crop variety, and so on.

47. In general, patterns of government intervention suffer three basic defects. First, the criteria that underlie the planning of these interventions lack an ecological orientation and are often dominated by short-term considerations. These criteria should discourage environmentally unsound farm practices and encourage farmers to maintain and improve their soils, forests, and waters.

48. The second defect is that agricultural policy tends to operate within a national framework with uniform prices and subsidies, standardized criteria for the provision of support services, indiscriminate financing of infrastructure investments, and so forth. Policies that vary from region to region are needed to reflect different regional needs, encouraging farmers to adopt practices that are ecologically sustainable in their own areas.

> The problem in agriculture is not faceless. I as a farmer am a potential victim of the system that we now operate under. Why are approximately a quarter of Canadian farmers facing the immediate prospects of farm bankruptcy? It is directly related to the general concept of a cheap food policy that has constituted a cornerstone of federal agricultural policy since the beginning of settlement.
>
> We regard the current cheap food policy as a form of economic violence that is contributing towards soil exploitation and the growing impersonal relationship between farmers and the soil for economic survival. It is a policy of industrialization that can lead only towards disaster economically for us as farmers, and environmentally for us all as Canadians and as world citizens.
>
> Wayne Easter
> President, National Farmers' Union
> WCED Public Hearing
> Ottawa, 26/27 May 1986

49. The importance of regional policy differentiation can be easily illustrated:

- Hill areas may require incentive prices for fruits and subsidized supplies of foodgrains to induce farmers to shift towards horticulture, which may be ecologically more sustainable.

- In areas prone to wind and water erosion, public intervention through subsidies and other measures should encourage farmers to conserve soil and water.

- Farmers on land over recharge areas for underground aquifers subject to nitrate pollution might be given incentives to maintain soil fertility and increase productivity by means other than nitrate fertilizers.

50. The third defect in government intervention lies in incentive structures. In industrialized countries, overprotection of farmers and overproduction represent the accumulated result of tax reliefs, direct subsidies, and price controls. Such policies are now studded with contradictions that encourage the degradation of the agricultural resource base and. in the long run, do more harm than good to the agricultural industry. Some governments now recognize this and are making efforts to change the focus of the subsidies from production growth to conservation.

51. On the other hand, in most developing countries the incentive structure is weak. Market interventions are often ineffective for lack of an organizational structure for procurement and distribution. Farmers are exposed to a high degree of uncertainty, and price support systems have often favoured the urban dweller or are limited to a few commercial crops, leading to distortions of cropping patterns that add to destructive pressures on the resource base. In some cases, price controls reduce the incentive to produce. What is required, in many cases, is nothing less than a radical attempt to turn the 'terms of trade' in favour of farmers through pricing policy and government expenditure reallocation.

52. Strengthening food security from a global point of view requires reducing incentives that force overproduction and non-competitive production in the developed market economies and enhancing those that encourage food production in developing countries. At the same time, these incentive structures must be redesigned to promote farming practices that conserve and enhance the agricultural resource base.

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2. A Global Perspective

53. Trade in agricultural products tripled between 1950 and 1970; it has doubled since then. Yet, when it comes to farming, countries are at their most conservative, continuing to think mainly in local or national terms and concerned, above all, to protect their own farmers at the expense of competitors.

54. Shifting food production towards food-deficit countries will require a major shift in trading patterns. Countries must recognize that all parties lose through protectionist barriers, which reduce trade in food products in which some nations may have genuine advantage. They must begin by redesigning their trade, tax, and incentive systems using criteria that include ecological and economic sustainability and international comparative advantage.

55. The incentive-driven surpluses in developed market economies increase pressures to export these surpluses at subsidized prices or as non-emergency food aid. Donor and receiving countries should be responsible for the impacts of aid and use it for long-term objectives. It can be beneficially used in projects to restore degraded lands, build up rural infrastructure, and raise the nutrition level of vulnerable groups.

3. The Resource Base

56. Agricultural production can only be sustained on a long-term basis if the land, water, and forests on which it is based are not degraded. As suggested, a reorientation of public intervention will provide a framework for this. But more specific policies that protect the resource base are needed to maintain and even enhance agricultural productivity and the livelihoods of all rural dwellers.

3.1 Land Use

57. The initial task in enhancing the resource base will be to delineate broad land categories:

- enhancement areas, which are capable of sustaining intensive cropping and higher population and consumption levels;

- prevention areas, which by common consent should not be developed for intensive agriculture or. where developed, should be converted to other uses; and

- restoration areas, where land stripped of vegetative cover has either totally lost its productivity or had it drastically reduced.

58. Identifying land according to 'best use' criteria requires information that is not always available. Most industrial nations possess inventories and descriptions of their lands, forests, and waters that are detailed enough to provide a basis for delineating land categories. Few developing countries have such inventories, but they can and should develop them quickly using satellite monitoring and other rapidly changing techniques./32

59. Selection of land for each category could be made the responsibility of a board or commission representing the interests involved, especially the poor and more marginalized segments of the population. The process must be public in character, with publicly agreed criteria that combine the best use approach with the level of development required to sustain livelihood. Classifying land according to best use will determine variations in infrastructure provision, support services, promotional measures, regulatory restrictions, fiscal subsidies, and other incentives and disincentives.

60. Lands identified as prevention areas should be denied supports and subsidies that would encourage their development for intensive agriculture. But such areas might well support certain ecologically and economically sustainable uses such as grazing. fuelwood plantations, fruit farming, and forestry. Those redesigning support systems and incentives should focus on a broader range of crops, including those that enhance grazing, soil and water conservation, and so on.

61. In vast areas today natural factors and land use practices have reduced productivity to a point too low to sustain even subsistence farming. Treatment of these areas must vary from site to site. Governments should give priority to establishing a national policy and multidisciplinary programmes and to creating or strengthening

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institutions to restore such areas. Where these already exist, they should be better coordinated and designed. The UN Plan of Action to Combat Desertification, which is already in place, requires more support, particularly financial.

62. Restoration may require limits on human activities so as to permit the regeneration of vegetation. This can be difficult where there are large herds of animals or large numbers of people. for the agreement and participation of the local people are of the highest importance. The state, with the cooperation of those living locally, could protect these areas by declaring them national reserves. Where these areas are privately held, the state might wish either to purchase the land from the owners or to provide incentives for its restoration.

Intensive agriculture may quickly exhaust the soil cover. causing its degradation, unless some special soil protection measures aimed at constant restoration and expanded reproduction of fertility are taken. The task of agriculture is thus not confined to obtaining the biological product but extends to constant maintenance and augmentation of soil fertility. Otherwise we will very quickly consume what by right belongs to our children, grandchildren, and great-grandchildren, to say nothing of more distant descendants.

It is this misgiving - that our generation lives to a certain extent at the expense of the coming generations, thoughtlessly drawing on the basic reserves of soil fertility accumulated in the millennia of the biospheric development. instead of living off the current annual increment - that causes the increasing concern of scientists dealing with the state of the planetary soil cover.

B. G. Rozanov
Moscow State University
WCED Public Hearing
Moscow, 11 Dec 1986

3.2 Water Management

63. Improvements in water management are essential to raise agricultural productivity and to reduce land degradation and water pollution. Critical issues concern the design of irrigation projects and the efficiency of water use.

64. Where water is scarce, an irrigation project should maximize productivity per unit of water; where water is plentiful, it must maximize productivity per unit of land. But local conditions will dictate how much water can be used without damaging the soil. Salinization. alkalization, and waterlogging can be avoided by a more careful approach to drainage, maintenance, cropping patterns. the regulation of water quantities, and more rational water charges. Many of these objectives will be easier to realize in small-scale irrigation projects. But whether small or large, the projects must be designed with the abilities and aims of the participating farmers in mind, and then involve them in the management.

65. In some areas excessive use of ground-water is rapidly lowering the water table - usually a case where private benefits are being realized at society's expense. Where ground-water use exceeds the recharge capacity of local aquifers, regulatory or fiscal controls become essential. The combined use of ground and surface water can improve the timing of water availability and stretch limited supplies.

3.3 Alternatives to Chemicals

66. Many countries can and should increase yields by greater use of chemical fertilizers and pesticides, particularly in the developing world. But countries can also improve yields by helping farmers to use organic nutrients more efficiently. Hence governments must encourage the use of more organic plant nutrients to complement chemicals. Pest control must also be based increasingly on the use of natural methods. (See Box 5-2) These strategies require changes in public policies, which now encourage the increased use of chemical pesticides and fertilizers. The legislative, policy, and research capacity for advancing non-chemical and less-chemical strategies must be established and sustained.

67. Chemical fertilisers and pesticides are heavily subsidized in many countries. These subsidies promote chemical use precisely in the more commercially oriented agricultural areas where their environmental damage may already outweigh any increases in productivity they bring. Hence different regions will require different policies to regulate and promote chemical use.

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68. Legislative and institutional frameworks for controlling agrochemicals must be greatly strengthened everywhere.

69, Industrialized countries must tighten controls on pesticide exports. (See Chapter 8.) Developing countries must possess the basic legislative and institutional instruments to manage the use of agricultural chemicals within their countries. And they will need technical and financial assistance to do so.

3.4 Forestry and Agriculture

69. Undisturbed forests protect watersheds, reduce erosion. offer habitats for wild species, and play key roles in climatic systems. They are also an economic resource providing timber, fuelwood, and other products. The crucial task is to balance the need to exploit forests against the need to preserve them.

70. Sound forest policies can be based only on an analysis of the capacity of the forests and the land under them to perform various functions. Such an analysis might lead to some forests being cleared for intensive cultivation, others for livestock; some forestland might be managed for increased timber production or agroforestry use and some left intact for watershed protection, recreation, or species conservation. The extension of agriculture into forest areas must be based on scientific classification of land capacities.

71. Programmes to preserve forest resources must start with the local people who are both victims and agents of destruction, and who will bear the burden of any new management scheme./33 They should be at the centre of integrated forest management, which is the basis of sustainable agriculture.

72. Such an approach would entail changes in the way governments set development priorities, as well as the evolution of greater responsibility to local governments and communities. Contracts covering forest use will have to be negotiated, or overall environmental sustainability of forest exploitation and overall environmental and ecosystem conservation. Prices for forest products need to reflect the true resource value of the goods.

Box 5-2

Natural Systems of Nutrient Supply and Peat Control

- Crop residues and farmyard manure are potential sources of soil nutrients.
- Organic wastes reduce run-off, increase the take-up of other nutrients, and improve soil's water-holding and erosion-resistance capacity.
- Using farmyard manure, especially in conjunction with intercropping and crop rotation, can greatly lower production costs.
- Overall systems efficiency is enhanced if manure or vegetable biomass is anaerobically digested in biogas plants, yielding energy for cooking and to run pumps, motors, or electric generators.
- Natural systems of biological nitrogen fixation through the use of certain annual plants, trees, and micro-organisms have a high potential.
- Integrated pest management (IPM) reduces the need for agrochemicals. improves a country's balance of payments. releases foreign exchange for other development projects, and creates jobs where they are most needed.
- IPM requires detailed information about pests and their natural enemies, seed varieties tailored to resist pests, integrated cropping patterns, and farmers who support the approach and are willing to modify farm practices to adopt it.

73. Portions of forests may be designated as prevention areas. These are predominantly national parks, which could be set aside from agricultural exploitation to conserve soil, water, and wildlife. They may also include marginal lands whose exploitation accelerates land degradation through erosion or desertification. In this connection, the reforestation of degraded forest areas is of utmost importance. Conservation areas or national parks can also conserve genetic resources in their natural habitats. (See Chapter 6.)

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74. Forestry can also be extended into agriculture. Farmers can use agroforestry systems to produce food and fuel. In such systems, one or more tree crops are combined with one or more food crops or animal farming on the same land, though sometimes at different times. Well-chosen crops reinforce each other and yield more food and fuel than when grown separately. The technology is particularly suitable for small farmers and for poor-quality lands. Agroforestry has been practised by traditional farmers everywhere. The challenge today is to revive the old methods, improve them, adapt them to the new conditions. and develop new ones./34

75. International forestry research organizations should work in various tropical countries in various ecosystems along the lines now followed by the Consultative Group on International Agricultural Research. There is considerable scope for institution building and additional research on forestry's role in agricultural production, for example by developing models that better predict the effects on water and soil loss of removing specific portions of forest cover.

3.5 Aquaculture

76. Fisheries and aquaculture are critical to food security in that they provide both protein and employment. The greater part of world fish supply comes from marine fisheries, which yielded 76.8 million tons in 1983. Landings have increased by 1 million tons per year over the past few years; by the end of the century, a catch of around 100 million tons should be possible./35 This is well short of the projected demand. There are indications that much of the naturally available freshwater fish stocks are fully exploited or damaged by pollution.

77. Aquaculture. or 'fish-farming', which differs from conventional fishing in that fish are deliberately reared in controlled water bodies, can help meet future needs. Yields from aquaculture have doubled during the last decade and now represent about 10 per cent of world production of fishery products./36 A five- to tenfold increase is projected by the year 2000, given the necessary scientific, financial, and organizational support./37 Aquaculture can be undertaken in paddy fields, abandoned raining excavations, small ponds, and many other areas with some water, as well as on various commercial scales: individual, family, cooperative, or corporate. The expansion of aquaculture should be given high priority in developing and developed countries.

4. Productivity and yields

78. The conservation and enhancement of agriculture's resource base will increase production and productivity. But specific measures are required to make inputs more effective. This is best done by strengthening the technological and human resource base for agriculture in developing countries.

> Thus at the root of this environmental problem is a land problem that has to be solved if any serious ecological policy is to be taken - and reorientation of the agricultural policy has to be undertaken. I believe that any conservationist policy has to be followed by a coherent agricultural policy that will meet the need not only of preservation as such but also meet the needs of the Brazilian population.
>
> Julio M.G. Gaiger
> President, National Indian Support Association
> WCED Public Hearing
> Sao Paulo. 28/29 Oct 1985

4.1 The Technological Base

Blends of traditional and modern technologies offer possibilities for improving nutrition and increasing rural employment on a sustainable basis. Biotechnology, including tissue culture techniques, technologies for preparing value-added products from biomass, micro-electronics, computer sciences, satellite imagery, and communication. technology are all aspects of frontier technologies that can improve agricultural productivity and resource management./38

80. Providing sustainable livelihoods for resource-poor farmers presents a special challenge for agricultural research. The major advances in agricultural technology in recent decades are better suited to stable, uniform, resource-rich conditions with good soils and ample water supplies. New technologies are most urgently needed in sub-Saharan Africa and the remoter areas of Asia and Latin America, which typically have unreliable rainfall, uneven topography, and poorer soils, and hence are unsuited to Green Revolution technologies.

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81. To serve agriculture in these areas, research has to be less centralized and more sensitive to farmers' conditions and priorities. Scientists will need to start talking to poor farmers and basing research priorities on growers' priorities. Researchers must learn from and develop the innovations of farmers and not just the reverse. More adaptive research should be done right on the farm, using research stations for referral and with farmers eventually evaluating the results.

82. Commercial enterprises can help develop and diffuse technology, but public institutions must provide the essential framework for agricultural research and extension. Pew academic and research institutions in developing regions are adequately funded. The problem is most acute in the low-income countries, where expenditure on agricultural research and extension amounts to 0.9 per cent of total agricultural income, as against 1.5 per cent in the middle-income countries./39 Research and extension efforts must be greatly expanded, especially in areas where climate, soils, and terrain pose special problems.

83. These areas particularly will need new seed varieties, but so will much developing-country agriculture. At present, 55 per cent of the world's scientifically stored plant genetic resources is controlled by institutions in industrial countries, 31 per cent by institutions in developing countries, and 14 per cent by International Agricultural Research centres./40 Much of this genetic material originated in developing countries. These gene banks must increase their inventories of material, improve their storage techniques, and ensure that the resources are readily accessible to research centres in developing countries.

84. Private companies increasingly seek proprietary rights to improved seed varieties, often without recognizing the rights of the countries from which the plant matter was obtained. This could discourage countries rich in genetic resources from making these internationally available and thus reduce the options for seed development in all countries. The genetic research capabilities of developing countries are so limited that agriculture there could become excessively dependent on private gene banks and seed companies elsewhere. Thus international cooperation and a clear understanding on the sharing of gains are vital in critical areas of agricultural technology, such as the development of new seed varieties.

4.2 Human Resources

85. The technological transformation of traditional agriculture will be difficult without a matching effort to develop human resources. (See Chapter 4.) This means educational reforms to produce researchers more attuned to the needs of rural peoples and agriculture. Illiteracy is still widespread among the rural poor. But efforts to promote literacy should focus attention on functional literacy covering the efficient use of land, water, and forests.

86. Despite women's critical role in agriculture, their access to education and their representation in research, extension, and other support services is woefully inadequate. Women should be given the same educational opportunities as men. There should be more female extension workers, and women should participate in field visits. Women should be given more power to take decisions regarding agricultural and forestry programmes.

4.3 Productivity of Inputs

87. In traditional agriculture, local organic material provided farmers with sources of energy, nutrients, and ways of controlling pests. Today, these needs are increasingly met by electricity, petroleum products, chemical fertilizers, and pesticides. The cost of these inputs forms a growing proportion of agricultural costs, and wasteful use does economic and ecological harm.

88. One of the most important energy-related needs is mechanical power for irrigation. The efficiency of pumps could be greatly improved by providing appropriate incentives for equipment producers and farmers, and through effective extension work. Energy for irrigation pumps can also be provided by wind generators or by conventional internal combustion engines running on biogas produced from local biomass wastes. Solar dryers and solar coolers can save agricultural products. These non-conventional sources should be promoted, particularly in areas poor in energy resources.

89. Nutrients are lost when fertilizers are improperly applied. Often they leach away with the flow of water in a field and degrade local water supplies. Similar problems of waste and destructive side effects occur in the use

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of pesticides. Hence extension systems and chemical manufacturers will need to give priority to programmes to promote careful and economical use of these expensive, toxic materials.

5. Equity

90. The challenge of sustainable agriculture is to raise not just average productivity and incomes, but also the productivity and incomes of those poor in resources. And food security is not just a question of raising food production, but of ensuring that the rural and urban poor do not go hungry during the short term or midst a local food scarcity. All this requires the systematic promotion of equity in food production and distribution.

5.1 Land Reforms

91. In many countries where land is very unequally distributed land reform is a basic requirement. Without it, institutional and policy changes meant to protect the resource base can actually promote inequalities by shutting the poor off from resources and by favouring those with large farms, who are better able to obtain the limited credit and services available. By leaving hundreds of millions without options, such changes can have the opposite of their intended effect, ensuring the continued violation of ecological imperatives.

92. Given institutional and ecological variations, a universal approach to land reform is impossible. Each country should work out its own programme of land reform to assist the land-poor and to provide a base for coordinated resource conservation. The redistribution of land is particularly important where large estates and vast numbers of the land-poor coexist. Crucial components include the reform of tenancy arrangements, security of tenure, and the clear recording of land rights. In agrarian reforms the productivity of the land and. in forest areas, the protection of forests should be a major concern.

93. In areas where holdings are fragmented into many non-contiguous plots, land consolidation can ease the implementation of resource conservation measures. Promoting cooperative efforts by small farmers - in pest control or water management, for instance - would also help conserve resources.

94. In many countries women do not have direct land rights; titles go to men only. In the interests of food security, land reforms should recognize women's role in growing food. Women, especially those heading households, should be given direct land rights.

5.2 Subsistence Farmers and Pastoralists

95. Subsistence farmers, pastoralists, and nomads threaten the environmental resource base when processes beyond their control squeeze their numbers onto land or into areas that cannot support them.

96. The traditional rights of subsistence farmers, particularly shifting cultivators, pastoralists. and nomads, must therefore be protected from encroachments. Land tenure rights and communal rights in particular must be respected. When their traditional practices threaten the resource base, their rights may have to be curtailed, but only when alternatives have been provided. Most of these groups will need to be helped to diversify their livelihoods by entering the market economy through employment programmes and some cash-crop production.

97. Research should give early attention to the varied requirements of the mixed farming typical in subsistence agriculture. Extension and input supply systems must become more mobile to reach shifting cultivators and nomads and priority given to public investment to improve their cropland, grazing areas, and water sources.

5.3 Integrated Rural Development

98. Rural populations will continue to increase in many countries. With existing patterns of land distribution, the number of smallholders and landless households will increase by about 50 million, to nearly 220 million, by the year 2000./41 Together, these groups represent three-quarters of the agricultural households in developing countries./42 Without adequate livelihood opportunities, these resource-poor households will remain poor and be forced to overuse the resource base to survive.

99. Considerable effort has gone into creating strategies of integrated rural development, and the requirements and pitfalls are well known. Experience has shown that land reform is necessary but alone is not enough without support through the distribution of inputs and rural services. Smallholders, including - indeed especially

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- women, must be given preference when allocating scarce resources, staff, and credit. Small farmers must also be more involved in formulating agricultural policies.

100. Integrated rural development also requires resources to absorb the large increases in rural working populations expected in most developing countries through non-agricultural work opportunities, which should be promoted in rural areas. Successful agricultural development and the growth in incomes should open up opportunities in service activities and small-scale manufacturing if supported by public policy.

5.4 Food Availability Fluctuations

101. Environmental degradation can make food shortages more frequent and more severe. Hence sustainable agricultural development will reduce the season-to-season variability in food supplies. But such systems cannot eliminate it. There will be weather-induced fluctuations, and the growing dependence on only a few crop varieties over large areas may amplify the effects of weather and pest damage. Often it is the poorest households and the ecologically disadvantaged regions that suffer most from these shortages.

102. Food stocks are crucial in dealing with shortages. At present, the world stock of cereals is on the order of 20 per cent of annual consumption: The developing world controls about one-third of the stock and the industrial world, two-thirds. More than half the developing-country stock is in two countries - China and India. Stock levels in most of the others provide only for immediate operational requirements; there is little by way of a reserve./43

As agriculture production is being developed, a rising number of farmers have been able to purchase tractors. But they find that, after using them for a year, it becomes much more expensive than they expected because they have to spend a tremendous amount of money on expensive spare parts. Perhaps we might recommend that Indonesia establish a factory that makes these spare parts, before they continue encouraging introduction of tractors in agriculture.

For this reason, a number of loans that the government has been providing for farmers to modernize their agricultural techniques, particularly buying tractors, have not been paid back. If the tractors were still running, they could probably pay back their loans. In fact, now these tractors are becoming a problem themselves, because they sit around getting rusty. and thus turning into pollution.

Andi Mappasala
Chairman. Yayasan Tellung Poccoe
WCED Public Hearing
Jakarta. 26 March 1985

103. The food stocks of industrialized countries are essentially surpluses, and provide a basis for emergency assistance, which must be maintained. But emergency food aid is a precarious basis for food security: developing countries should build up national stocks in surplus years to provide reserves as well as encouraging development of food security at the household level. To do this, they will need an effective system of public support for measures facilitating the purchase, transportation, and distribution of food. The provision of strategically located storage facilities is critical both to reduce post-harvest losses and to provide a base for quick interventions in emergencies.

104. During most food shortages, poor households not only cannot produce food but also lose their usual sources of income and cannot buy the food that is available. Hence food security also requires that machinery is available promptly to put purchasing power in the hands of disaster-struck households, through emergency public works programme, and through measures to protect small farmers from crop failures.

V. Food for the Future

105. The challenge of increasing food production to keep pace with demand, while retaining the essential ecological integrity of production systems, is colossal both in its magnitude and complexity. But we have the Knowledge we need to conserve our land and water resources. New technologies provide opportunities for increasing productivity while reducing pressures on resources. A new generation of farmers combine experience with education. With these resources at our command, we can meet the needs of the human family. Standing in the way is the narrow focus of agricultural planning and policies.

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106. The application of the concept of sustainable development to the effort to ensure food security requires systematic attention to the renewal of natural resources. It requires a holistic approach focused on ecosystems at national, regional. and global levels, with coordinated land use and careful planning of water usage and forest exploitation. The goal of ecological security should be embedded firmly in the mandates of FAO. other UN organizations that deal with agriculture, and all other appropriate international agencies. It will also require an enhancement and reorientation of international assistance. (See Chapter 3.)

107. The agricultural systems that have been built up over the past few decades have contributed greatly to the alleviation of hunger and the raising of living standards. They have served their purposes up to a point. But they were built for the purposes of a smaller, more fragmented world. New realities reveal their inherent contradictions. These realities require agricultural systems that focus as much attention on people as they do on technology, as much on resources as on production, as much on the long term as on the short term. Only such systems can meet the challenge of the future.

Footnotes

1/ Based on data from FAO, Production Yearbook 1985 (Rome: 1986).

2/ Based on World Bank estimates for 1980, according to which 340 million people in developing countries (excluding China) did not have enough income to attain a minimum calorie standard that would prevent serious health risks and stunted growth in children, and 730 million were below a higher standard that would allow an active working life. See World Bank, Poverty and Hunger: Issues and Options for Food Security in Developing Countries (Washington. DC: 1986).

3/ FAO, Yearbook of Food and Agriculture Statistics, 1951 (Home: 1952); FAO, Production Yearbook 1985. op. cit.

4/ FAO, Yearbook of Food and Agricultural Statistics, Trade Volume, Part 2 1951 and Trade Yearbook 1982 and 1984 (Rome: 1952. 1983, and 1985).

5/ FAO, Trade Yearbook 1968 and Commodities Review and Outlook 1984-85 (Rome: 1969 and 1986).

6/ FAO, Yearbook of Food and Agricultural Statistics, Trade Volume, Part 2 1954 (Rome: 1955); FAO, Commodities Review, op. cit.

7/ FAO, Production Yearbook 1984 (Rome: 1985).

8/ L.R. Brown. 'Sustaining World Agriculture,' in L.R. Brown et al., State of the World 1987 (London: W.W. Norton. 1987).

9/ A. Gear (ed.), The Organic Food Guide (Essex: 1983).

10/ USSR Committee for the International Hydrological Decade, World Water Balance and Water Resources of the Earth (Paris: UNESCO, 1978).

11/ FAO, Yearbook of Food and Agricultural Statistics 1951 and Production Yearbook 1984. op. cit.

12/ 'Dairy, Prairie', The Economist, 15 November 1986.

13/ WCED Advisory Panel on Food Security, Agriculture, Forestry and Environment, Food Security (London: Zed Books. 1987).

14/ The term pesticides is used in a generic sense in this report and covers insecticides, herbicides. fungicides, and similar agricultural inputs.

15/ World Bank. World Development Report 1986 (New York: Oxford University Press. 1986).

16/ Brown, op. cit.

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17/ Standing Committee on Agriculture. Fisheries and Forestry. Soil at Risk: Canada's Eroding Future, A Report on Soil Conservation to the Senate of Canada (Ottawa: 1984).

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20/ FAO, Land, Food and People (Rome: 1984).

21/ I. Szabolcs. 'Agrarian Change', prepared for WCED, 1985.

22/ Gear. op. cit./

23/ J. Bandyopadhyay, 'Rehabilitation of Upland Watersheds', prepared for WCED, 1985.

24/ UNEP. 'General Assessment of Progress in the Implementation of the Plan of Action to Combat Desertification 1978-1984'. Nairobi, 1984; WCED Advisory Panel, op. cit.

25/ UNEP, op. cit.

26/ Ibid.

27/ Ibid.

28/ Ibid.

29/ Ibid.

30/ FAO, Agriculture Towards 2000 (Rome: 1981)

31/ FAO, Potential Population Supporting Capacities of Lands in the Developing World (Rome: 1982).

32/ The land capability classification developed by the U.S. Bureau of Land Management is an example of how the problem could be approached. A broader type of classification is implicit in FAO, Potential Population Supporting Capacities.

33/ INDERENA, Caguan-Cagueta Report (Bogota, Colombia: 1985).

34/ The agroforestry programmes implemented in India are examples of such an approach. They have been adopted enthusiastically by many farmers.

35/ FAO. World Food Report (Rome: 1985); WCED Advisory Panel, op. cit.

36/ WCED Advisory Panel, op. cit.

37/ Ibid.

38/ Ibid.

39/ FAO, World Food Report, op. cit.

40/ Data from Dag Hammarskjold Foundation, Sweden, in Centre for Science and Environment, op. cit.

41/ FAO estimates quoted in WCED Advisory Panel, op. cit.

42/ Ibid.

43/ FAO, Food Outlook (Rome: 1986).

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